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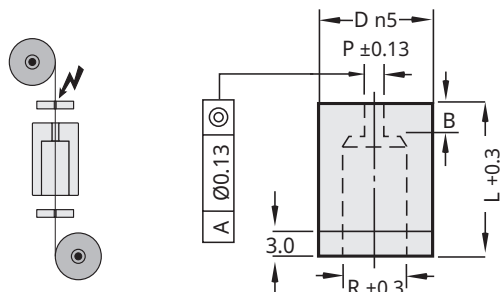
Punches and Retainers / Stamping



Code: **MKB**

Matrix with Counter-bore Relief

Blank type & with starting holes for wire-edm ISO 8977



With starting holes
for "wir e-edm".

D	P	B - point length		R	L
		S (standard)	A (optional)		
8	0.8	4	-	4	20
10	0.8	4	8	6	22
13	1.6	5	8	8	25
16	1.6	5	8	9.5	28
20	1.6	5	12	12	30
22	1.6	6	12	15	32
25	1.6	6	12	17.5	35
28	1.6	6	12	17.5	40
30	1.6	6	12	17.5	60
32	1.6	6	12	21	
35	1.6	6	12	21	
38	1.6	8	12	27	
40	1.6	8	12	27	
45	3.2	8	12	36	
50	3.2	8	12	41	



Order:
MKB. D x L - B

Order Example:
MKB. 20x25-S



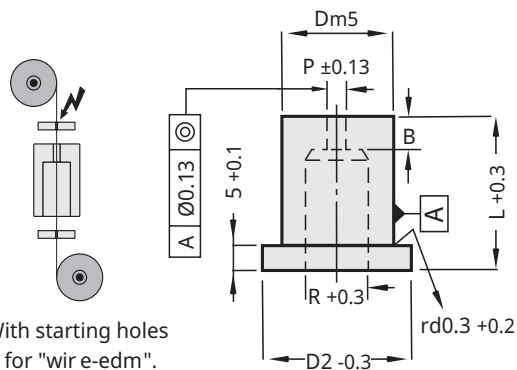
Material: 1.3343 (M2)
Hardness: 60 - 62 HRC



Code: **MLB**

Headed Matrix with Counter-bore Relief

Blank type & with starting holes for wire-edm ISO 8977



With starting holes
for "wir e-edm".

D	D2	P	B - point length		R	L
			S (standard)	A (optional)		
8	11	0.8	4	-	4	20
10	13	0.8	4	8	6	22
13	16	1.6	5	8	8	25
16	19	1.6	5	8	9.5	28
20	23	1.6	5	12	12	30
22	25	1.6	6	12	15	32
25	28	1.6	6	12	17.5	35
28	31	1.6	6	12	17.5	40
30	33	1.6	6	12	17.5	
32	35	1.6	6	12	21	
35	38	1.6	6	12	21	
38	41	1.6	8	12	27	
40	43	1.6	8	12	27	



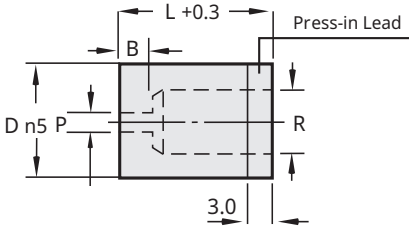
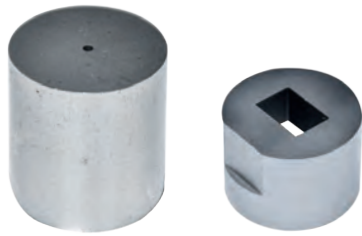
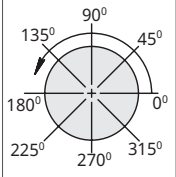
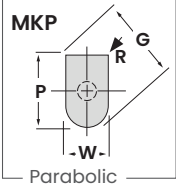
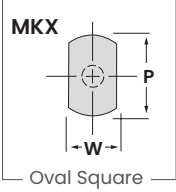
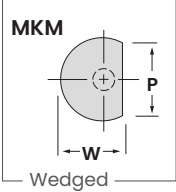
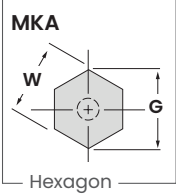
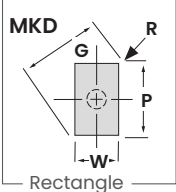
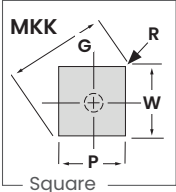
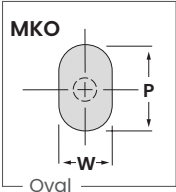
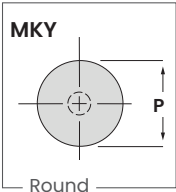
Order:
MLB. D x L - B

Order Example:
MLB. 16x20-S



Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

SHAPES



Code: **MK..**

Matrix - Straight ISO 8977

Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.

Please refer to the table on the left side
MKY - MKO - MKK - MKD - MKA - MKM - MKX - MKP

Code	D	B - point length		ROUND (MKY) Range P	SHAPE		R	L									
		S (standard)	A (optional)		Min. W	Max. P/G		20	22	25	28	30	32	35	40	60	
MK...08	8	4	-	1.50 - 3.20	1.50 - 3.20		4	•	•	•	•	•	•	•			
MK...10	10	4	8	1.60 - 5.00	1.60 - 5.00		6	•	•	•	•	•	•	•			
MK...13	13	5	8	1.80 - 7.20	1.80 - 7.20		8	•	•	•	•	•	•	•			
MK...16	16	5	8	5.00 - 8.80	2.50 - 8.80		9.5	•	•	•	•	•	•	•			
MK...20	20	5	12	5.50 - 11.00	3.20 - 11.00		12	•	•	•	•	•	•	•	•		
MK...22	22	6	12	7.50 - 14.00	4.00 - 14.00		15	•	•	•	•	•	•	•	•		
MK...25	25	6	12	9.50 - 16.50	4.80 - 16.50		17.5	•	•	•	•	•	•	•	•		
MK...28	28	6	12	11.50 - 19.50	4.80 - 20.00		17.5	•	•	•	•	•	•	•	•		
MK...30	30	6	12	11.50 - 20.00	4.80 - 20.00		17.5	•	•	•	•	•	•	•	•		
MK...32	32	6	12	13.00 - 20.00	5.50 - 20.00		21	•	•	•	•	•	•	•	•		
MK...35	35	6	12	13.00 - 23.00	5.50 - 23.00		21	•	•	•	•	•	•	•	•		
MK...38	38	8	12	16.00 - 26.00	6.40 - 26.00		27	•	•	•	•	•	•	•	•		
MK...40	40	8	12	16.00 - 26.00	6.40 - 26.00		27			•	•	•	•	•	•		
MK...45	45	8	12	19.00 - 35.00	8.00 - 35.00		36						•		•	•	
MK...50	50	8	12	22.00 - 40.00	9.00 - 40.00		41							•		•	•



Order:
MK (type) D x L x shape (P/W/R/G)



Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

MKO 25 - (A) 32 - P=13.0 W=10.5 - K1

Code D L Shape Keying Info

* "B" Point length

* If you want the Point Length to be Optional, add code (A).
No code is written in the standard.



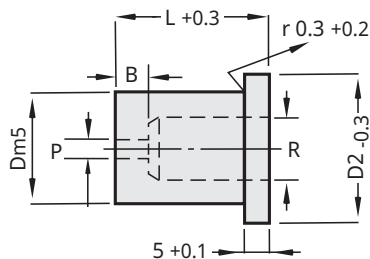
For details on Keying Information / Key Flat Options, and More Complex Special Shapes, please refer to the beginning of Category 6.

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Punches and Retainers / Stamping

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Punches and Retainers / Stamping



Code: **ML..**

Matrix - Headed ISO 8977

Apart from our standards, special dimensions and complex geometries can be produced according to your technical drawings.

Please refer to the table on the right side

MLY - MLO - MLK - MLD - MLA - MLM - MLX - MLP

Code	D	D2	B - point length		ROUND (MLY) Range P	SHAPE		R	L							
			S (standard)	A (optional)		Min. W	Max. P/G		20	22	25	28	30	32	35	40
ML...08	8	11	4	-	1.50 - 3.20	1.50 - 3.20	4	4	•	•	•	•	•	•	•	•
ML...10	10	13	4	8	1.60 - 5.00	1.60 - 5.00	6	6	•	•	•	•	•	•	•	•
ML...13	13	16	5	8	1.80 - 7.20	1.80 - 7.20	8	8	•	•	•	•	•	•	•	•
ML...16	16	19	5	8	5.00 - 8.80	2.50 - 8.80	9.5	9.5	•	•	•	•	•	•	•	•
ML...20	20	23	5	12	5.50 - 11.00	3.20 - 11.00	12	12	•	•	•	•	•	•	•	•
ML...22	22	25	6	12	7.50 - 14.00	4.00 - 14.00	15	15	•	•	•	•	•	•	•	•
ML...25	25	28	6	12	9.50 - 16.50	4.80 - 16.50	17.5	17.5	•	•	•	•	•	•	•	•
ML...28	28	31	6	12	11.50 - 19.50	4.80 - 20.00	17.5	17.5	•	•	•	•	•	•	•	•
ML...30	30	33	6	12	11.50 - 20.00	4.80 - 20.00	17.5	17.5	•	•	•	•	•	•	•	•
ML...32	32	35	6	12	13.00 - 20.00	5.50 - 20.00	21	21	•	•	•	•	•	•	•	•
ML...35	35	38	6	12	13.00 - 23.00	5.50 - 23.00	21	21	•	•	•	•	•	•	•	•
ML...38	38	41	8	12	16.00 - 26.00	6.40 - 26.00	27	27	•	•	•	•	•	•	•	•
ML...40	40	43	8	12	16.00 - 26.00	6.40 - 26.00	27	27	•	•	•	•	•	•	•	•



Order:

ML (type) D x L x **shape** (P/W/R/G)



Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

MLY 25 - (A) 32 - P=13.0

Code d2 L1 Shape

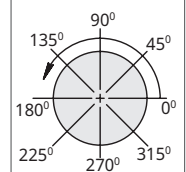
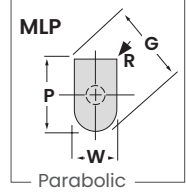
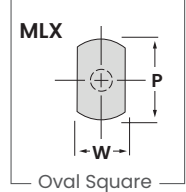
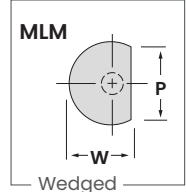
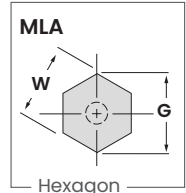
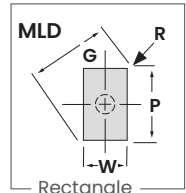
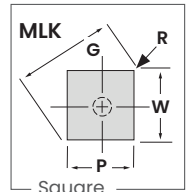
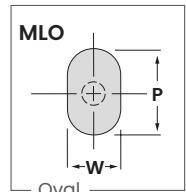
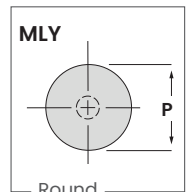
* "B" Point length

* If you want the Point Length to be Optional, add code (A).
No code is written in the standard.

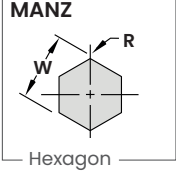
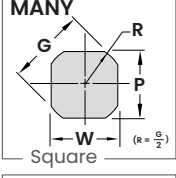
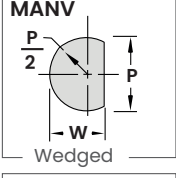
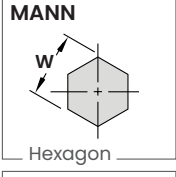
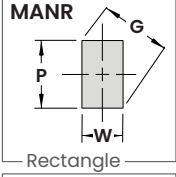
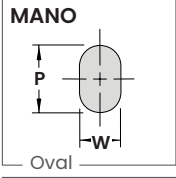
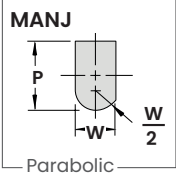
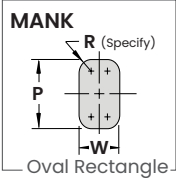
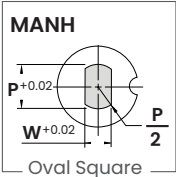
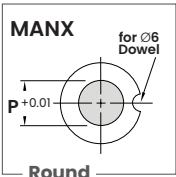


For details on Keying Information / Key Flat Options, and More Complex Special Shapes, please refer to the beginning of Category 6.

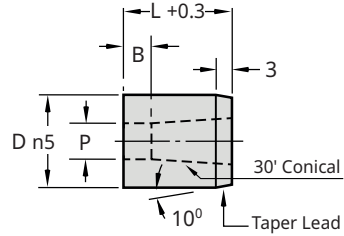
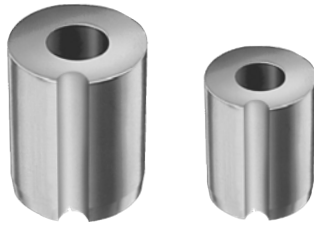
SHAPES



SHAPES



NEW



Code: **MAN..**

Matrix - Tapered Relief Ref.: NAAMS

This series of matrix comes standard with the 'PK6' Dowel Grooves (NAAMS standard) option for Ø6 dowels, unless otherwise specified.

Please refer to the table on the left side

MANX - MANH - MANK - MANJ - MANO - MANR - MANN - MANV - MANY - MANZ

Code	D	B - point length		ROUND	SHAPE		L											
		S (standard)	A (optional)	Range P	Min. W	Max. P/G	13	16	20	22	25	28	30	32	35	40		
MAN.. 10	10	4	5	1.60 - 6.80	1.30 - 6.80	●	●	●	●	●	●	●	●	●	●			
MAN.. 13	13	5	8	3.00 - 8.80	1.90 - 8.80	●	●	●	●	●	●	●	●	●	●			
MAN.. 16	16	5	8	7.40 - 10.80	1.90 - 10.80			●	●	●	●	●	●	●	●			
MAN.. 20	20	5	10	9.50 - 13.60	1.90 - 13.60			●	●	●	●	●	●	●	●			
MAN.. 22	22	6	10	10.50 - 15.00	1.90 - 15.00			●	●	●	●	●	●	●	●			
MAN.. 25	25	6	10	12.00 - 17.00	1.90 - 17.00			●	●	●	●	●	●	●	●			
MAN.. 32	32	6	12	16.00 - 22.00	1.90 - 22.00			●	●	●	●	●	●	●	●			
MAN.. 38	38	8	12	18.00 - 27.00	1.90 - 27.00			●	●	●	●	●	●	●	●			
MAN.. 40	40	8	12	18.00 - 27.00	1.90 - 27.00			●	●	●	●	●	●	●	●			
MAN.. 45	45	8	12	18.00 - 35.00	2.40 - 35.00				●	●	●	●	●	●	●	●		
MAN.. 50	50	8	12	18.00 - 40.00	4.00 - 40.00				●	●	●	●	●	●	●	●		



Order:
MAN (type) D x L x **shape** (P/W/G) x PK6

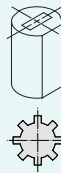


Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

MANX 25 - (A) 32 - P=13.0 - PK6
Code D L Shape Keying Info
* "B" Point length

* If you want the Point Length to be Optional, add code (A).
No code is written in the standard.



For details on Keying Information / Key Flat Options, and More Complex Special Shapes, please refer to the beginning of Category 6.

MAN matrixes conform to NAAMS™ standard for Tapered Relief Die Buttons.

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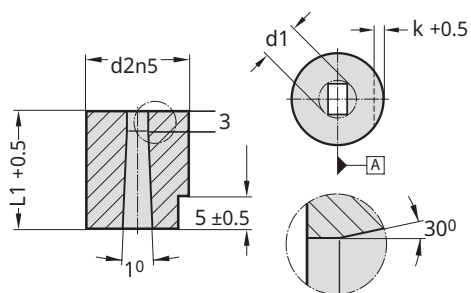
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Punches and Retainers / Stamping



Code: **MP..**

Matrix, Straight - Tapered Relief



Please refer to the table on the right side

MPY - MPO - MPK - MPD - MPA - MPM - MPX - MPP

d1 = shape	d2	k	L1
1.6 ~ 3.2	8	1	16
2 ~ 5	10		
3 ~ 7	13		
5 ~ 8	16	1.5	22
7 ~ 11	20		25
11 ~ 16	25		28
16 ~ 19	32	2.5	32
19 ~ 28	40		

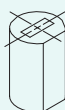
Order: **MP** (type) x d2 x L1
shape (P/W/R/G)



Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

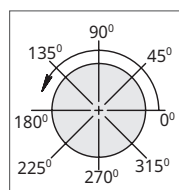
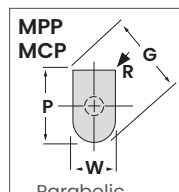
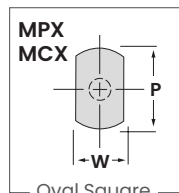
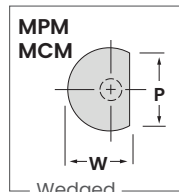
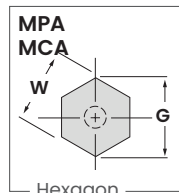
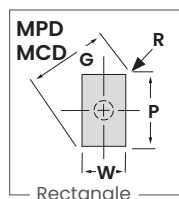
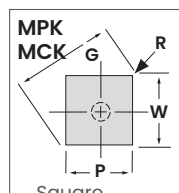
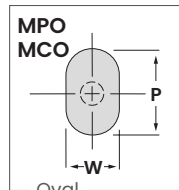
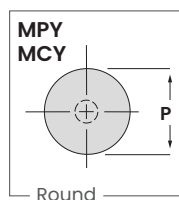
Order Example

MPX 25-28 P=16.0 W=12.5 - K1/90°
Code d2 L1 Shape Keying Info



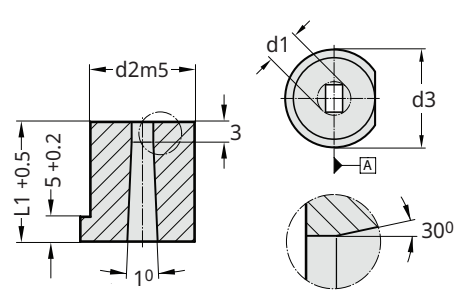
For details on Keying Information / Key Flat Options, please go to the beginning of Category-6.

SHAPES



Code: **MC..**

Matrix, Headed - Tapered Relief



Please refer to the table on the left side

MCY - MCO - MCK - MCD - MCA - MCM - MCX - MCP

d1 = shape	d2	d3	k	L1
1.6 ~ 3.2	8	11	1	16
2 ~ 5	10	13		
3 ~ 7	13	16		
5 ~ 8	16	19	1.5	22
7 ~ 11	20	23		25
11 ~ 16	25	28		28
16 ~ 19	32	35	2.5	32
19 ~ 28	40	43		

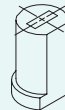
Order: **MC** (type) x d2 x L1
shape (P/W/R/G)



Material: 1.3343 (M2)
Hardness: 60 - 62 HRC

Order Example

MCY 16-25 P=8.0
Code d2 L1 Shape

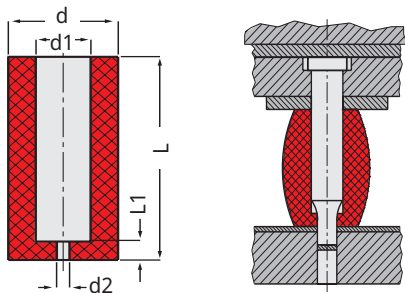


For details on Keying Information / Key Flat Options, please go to the beginning of Category-6.



Polyurethane Punch Stripper
Overspread on punch (shock absorber)

Code: **PYB**



Order Code	d	d1	d2	L1	L	Punch Length
PYB 445	4	17	1.6	5	45	56 / 63
PYB 655	6	19				
PYB 855	8	21	3	5	55	63
PYB 1055	10	23				71
PYB 1355	13	26				80
PYB 1655	16	30				90
PYB 2055	20	38	Punch Hole			100
PYB 2555	25	50				

d2: (1.6 - 3.0 mm) (d1) while opening hole diameter / drilling polyurethane (punch) bush should be applied and drilled in pressed (S max) position. Spring load is obtained while extending outwards.

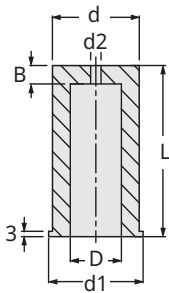
In S. max flexion, load coefficient data daN / Kg.

d1	17	19	21	23	26	30	38	50
S daN 3 mm	60	65	70	90	110	140	210	370
S daN 6 mm	115	120	130	160	190	230	360	650
S daN 9 mm	-	180	210	240	300	370	550	1020

S max: Load coefficient daN = Kg. / (10 newton), while polyurethane bush S max in pressed, load data is advisory.



In dies that polyurethane punch stripper is used, there is no need to dismantle stripper plate to make repair whetting and modification on die components, there is not any effect on precise parts, it is excellent for all painted / anodized, plastic plated and polished parts. It is compatible to use with oil and grease. It is overspread on punch. Placement is done according to the stripper hardness. There is no need for extra holder. Stepped-punch hole will be opened at the first stroke of press on stripper edge. **Especially, in large dies requiring very wide stripper plate, this product is very compatible.**



Heavy Duty

Light Duty

Code: **MTS**

Polyurethane Stripper for Ball Lock

Exchangeable and compatible with NAAMS standards.

Punch Dia.	D	d	L	d1	d2	B
10	9.75	18	44	21	1.6	6
			54			
			64			
			74			
13	12.75	32	44	26	3	6
			54			
			64			
			74			
16	15.75	28	44	31	3	6
			54			
			64			
			74			
20	19.75	33	44	36	3	7
			54			
			64			
			74			
25	24.75	40	44	43	3	7
			54			
			64			
			74			
32	31.70	50	44	55	3	7
			54			
			64			
			74			
40	39.70	60	44	65	3	8
			64			



Order:
MTS. d x L



Material:
Polyurethane / 95 Shore-A

Polyurethane Length for Strippers:

While ordering, please specify punch length.

Heavy Duty Ball Lock Punch Length	Ligh Duty Ball Lock Punch Length	Recommended Stripper Length "L "
80	71	44
90	80	54
100	90	64
110	100	74

NEW

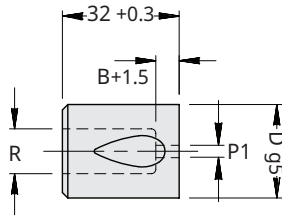


Ball Lock Matrix - Blank

Blank Type

Material:
1.3343 (M2)

Hardness:
60 - 62 HRC

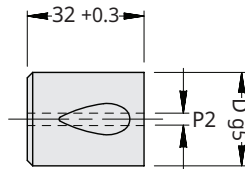


Code: **MBC**

Order Code	D	B	R	P1
MBC 13	13	4	5.8	1.2
MBC 16	16	5	8	1.6
MBC 20	20	5	11.9	2
MBC 25	25	6	16	3.6
MBC 32	32	6	20	4.4

Material:
1.3343 (M2)

Hardness:
60 - 62 HRC



Code: **MBE**

Order Code	D	P2
MBE 13	13	1.2
MBE 16	16	1.6
MBE 20	20	2
MBE 25	25	2.4
MBE 32	32	2.4

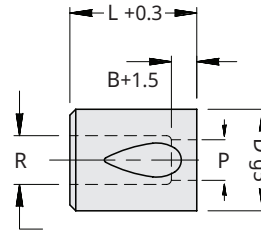
NEW



Ball Lock Matrix

Automotive Standard / Shaped Types

Code: **MB..**



Please refer to the table on the right side

MBY - MBO - MBK - MBD - MBA - MBM - MBX - MBP

Code	D	B	R	L	ROUND (MBY)	SHAPE	
					Range P	Min. W	Max. G/P
MB.. 13	13	4	6	32	1.6 - 5	1.6	5
MB.. 16	16	5	8	32	3.2 - 7.2	2	7.2
MB.. 20	20	5	12	32	4 - 11	2.5	11
MB.. 25	25	6	16	32	8 - 15	4	15
MB.. 32	32	6	20	32	11 - 19	5	19



Order:

MB. (type) D x L x Shape (P/W/G)

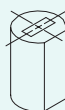
Material: 1.3343 (M2)

Hardness: 60 - 62 HRC

Order Example

MBY 25-32 P=13.0

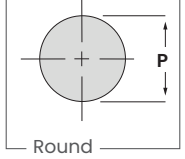
Code D L Shape



For details on Keying Information / Key Flat Options, please go to the beginning of Category-6.

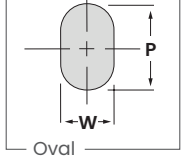
SHAPES

MBY



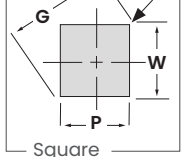
Round

MBO



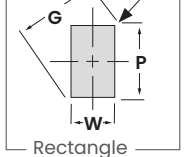
Oval

MBK



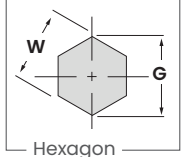
Square

MBD



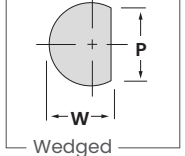
Rectangle

MBA



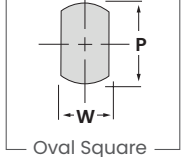
Hexagon

MBM



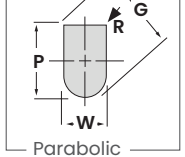
Wedged

MBX



Oval Square

MBP



Parabolic

